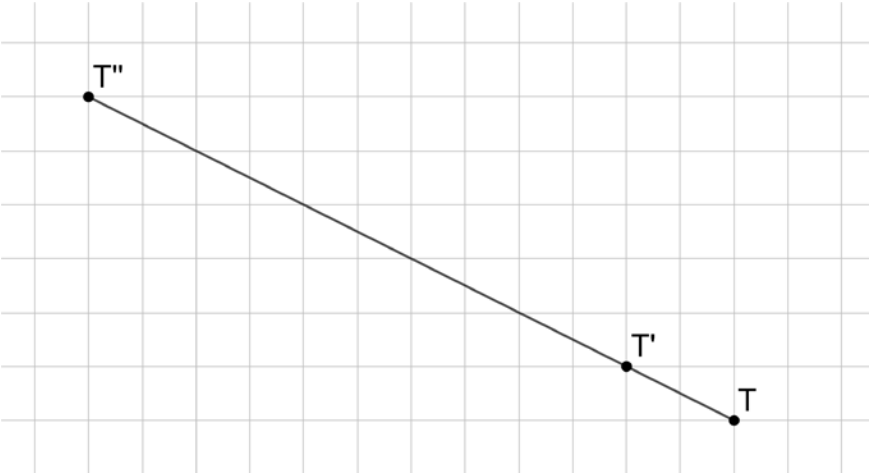


Points T , T' , and T'' are shown.



- 1. Determine the number of copies of $\overline{T'T'}$ that will fit on $\overline{T'T''}$.
- 2. Use the definition of dilation to explain why $\overline{T'T''}$ is a dilation of $\overline{T'T'}$.
- 3. Complete the statement.

Point

0	T
0	T'
0	T''

 partitions

0	$\overline{T'T'}$
0	$\overline{T'T''}$
0	$\overline{T'T''}$

 so that the ratio of

0	$\frac{T'T'}{T'T''}$
0	$\frac{T'T'}{T'T''}$

 is

0	$\frac{1}{5}$
0	$\frac{1}{6}$

Use the Pythagorean theorem to complete the table.

	Segment	Length
4.	$\overline{T'T'}$	
5.	$\overline{T'T''}$	
6.	$\overline{T'T''}$	

7. Complete the statements.

Point T' is

0

$\frac{1}{5}$

0

$\frac{1}{6}$

 of the distance from T to T'' . The scale factor that can be used to

multiply

0

$\overline{TT'}$

0

$\overline{T'T''}$

0

$\overline{TT''}$

0

$\overline{TT'}$

0

$\overline{T'T''}$

0

 $\overline{TT''}$

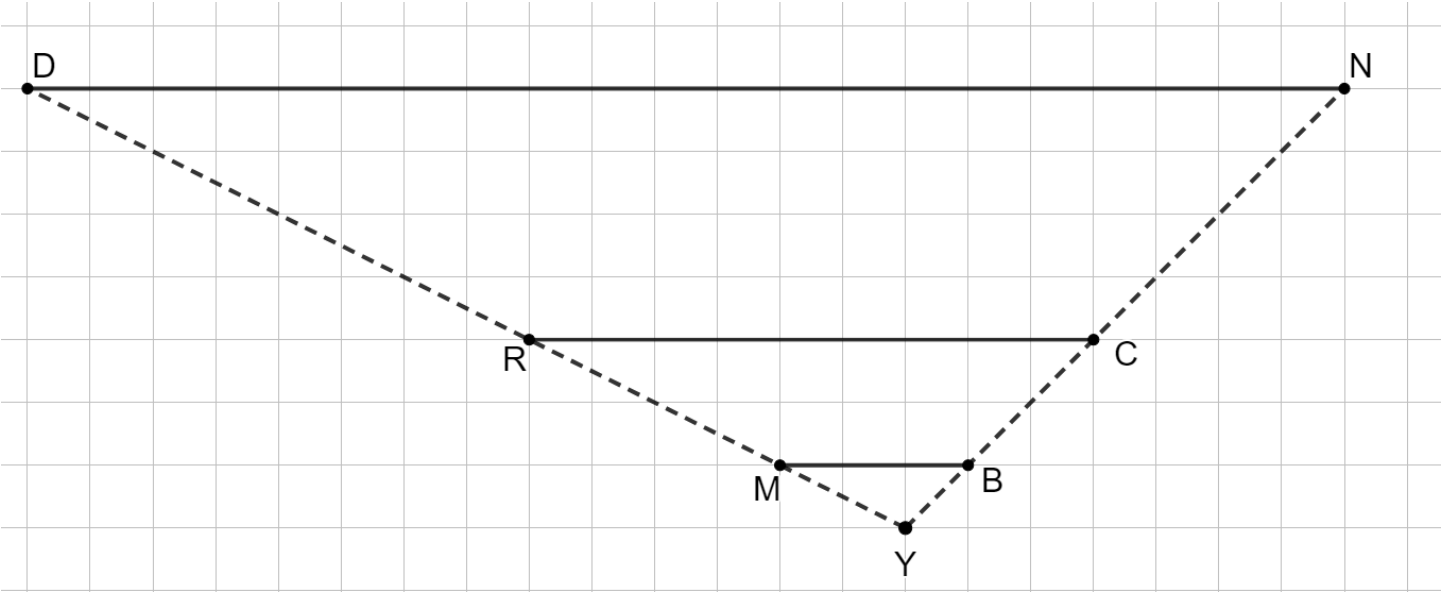
0

$\frac{1}{5}$

0

$\frac{1}{6}$

Line segments MB , RC , and DN are shown where $\overline{MB}$ is dilated using point Y to create $\overline{RC}$ and $\overline{DN}$.



Determine each of the ratios.

	Ratio	Length
8.	$YM:YR$	
9.	$MB:RC$	
10.	$MB:DN$	